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SEC-A
SUBJECT-Flexi Credit

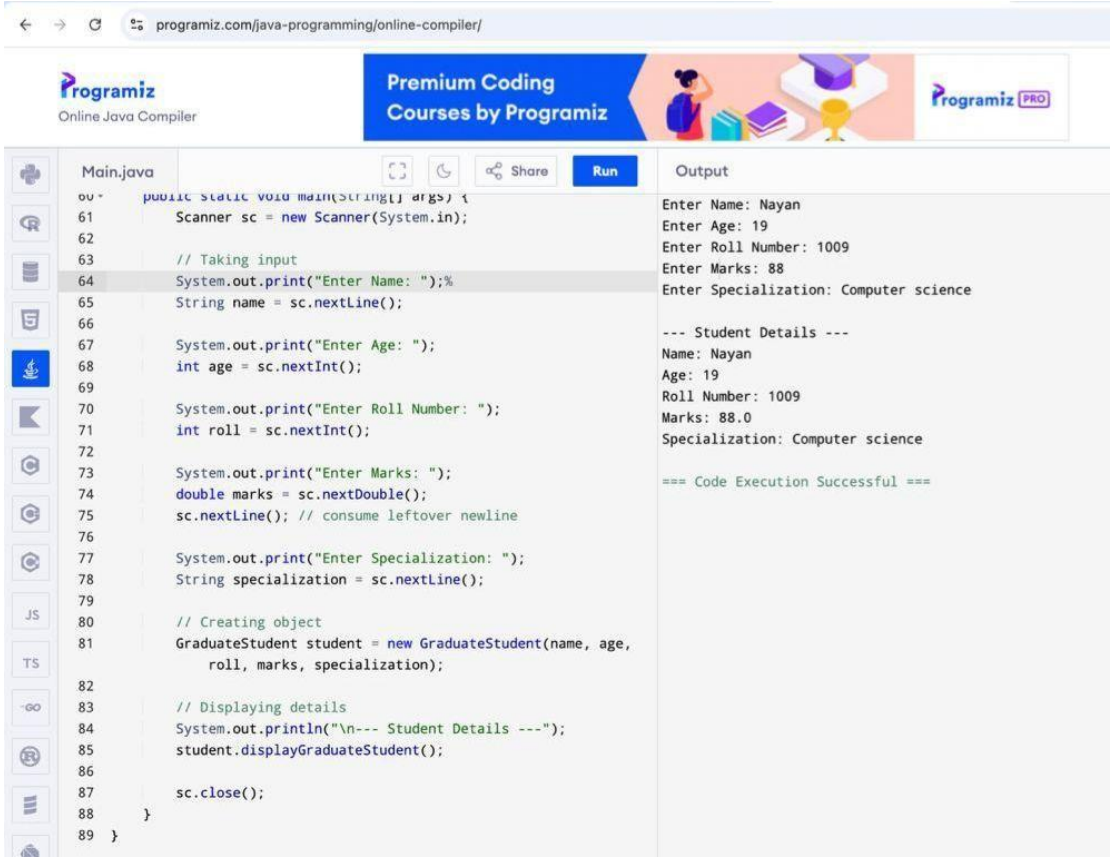
- **Problem Statement**

Student Management System Problem
Definition :
Design a Java program to manage student information using inheritance. Create a base class Person that stores common details like name and age. Derive a class Student that includes additional attributes such as roll number and marks. Further extend it into another class GraduateStudent with specialization details. The program should display complete student information using inherited properties.

Code-

```
import java.util.Scanner;

// Base class
class Person {
String name; int
age;

    // Constructor
    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    // Display method
    void displayPerson() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}

// Derived class class Student
extends Person {
```

```
    int rollNumber;
    double marks; //
    Constructor
    Student(String name, int age, int rollNumber, double marks) {
        super(name, age); this.rollNumber = rollNumber;
        this.marks = marks;
    }

    // Display method void
    displayStudent() {
    displayPerson();
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("Marks: " + marks);
    }
}

// Further derived class
class GraduateStudent extends Student {
    String specialization;

    // Constructor
    GraduateStudent(String name, int age, int rollNumber, double marks,
String specialization) { super(name, age,
        rollNumber, marks);
        this.specialization = specialization;
    }

    // Display method void
    displayGraduateStudent() {
    displayStudent();
        System.out.println("Specialization: " + specialization);
    }
}

// Main class
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Taking input
        System.out.print("Enter Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Age: ");
        int age = sc.nextInt();
```

```
        System.out.print("Enter Roll Number: ");
        int roll = sc.nextInt();
        System.out.print("Enter Marks: "); double
        marks = sc.nextDouble(); sc.nextLine(); //
        consume leftover newline

        System.out.print("Enter Specialization: ");
        String specialization = sc.nextLine();

        // Creating object
        GraduateStudent student = new GraduateStudent(name, age, roll, marks, specialization);

        // Displaying details
        System.out.println("\n--- Student Details ---");
        student.displayGraduateStudent();

        sc.close();
    }
}
```

OUTPUT-